

Discrete Mathematics (MATH 208)

Fall 2022

Friday, 10/07/2022

Homework 6: Chapters 13, 14, 15

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| Score: / 36 |
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Name (Print): \_\_\_\_\_

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## Chapter 13 - Sequences and Summation

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2 pts

1. A sequence begins 1, 3, 9, 15. Could it be an arithmetic sequence? Could it be a geometric sequence?

2 pts

2. What is the  $20^{th}$  term of the arithmetic sequence with initial term 4 and common difference 5?

2 pts

3. The  $8^{th}$  term of an arithmetic sequence is 20 and the  $12^{th}$  term is 40. What is the  $25^{th}$  term?

2 pts

4. What is the  $7^{th}$  term of the geometric sequence with initial term 3 and common ratio 4?

2 pts 5. Evaluate  $\sum_{j=1}^4 (j+1)^2$ .

2 pts 6. What is the sum of the first four thousand terms of the geometric sequence with initial term 3 and common ratio  $-1$ ?

## Chapter 14 - Recursively Defined Sequences

3 pts 1. List the first five terms of the sequence defined recursively by  $a_1 = 2$ , and, for  $n \geq 2$ ,  
 $a_n = a_{n-1}^2 - 1$ .

3 pts 2. Let  $r$  be a fixed real number different from 0. For a positive integer  $n$ , the symbol  $r^n$  means the product of  $n$   $r$ 's. For convenience,  $r^0$  is defined to be 1. Give a recursive definition of  $r^n$  analogous to the definition of  $n!$  given in this chapter.

3 pts

3. Give a recursive definition of the geometric sequence with initial term 3 and common ratio 2.

3 pts

4. Give a recursive definition of the geometric sequence with initial term  $a$  and common ratio  $r$ .

## Chapter 15 - Recursively Defined Sets

3 pts

1. A set,  $S$ , of strings over the alphabet  $\Sigma = \{a, b, c\}$  is defined recursively by:

- (i)  $a \in S$ ,
- (ii) if  $x \in S$  then  $bx \in S$ .

List all the strings in  $S$  of length seven or less.

3 pts

2. A set,  $S$ , of positive integers is defined recursively by the rule:

(i)  $1 \in S$ ,

(ii) If  $n \in S$ , then  $2n - 1 \in S$ .

List all the elements in the set  $S$ .

3 pts

3. Give a recursive definition of the set of positive integers that end with the digit 1.

3 pts

4. Give a recursive definition of the set of strings over the alphabet  $\Sigma = \{a, b, c\}$  of the form  $aaa \cdots abccc \cdots c$ . More carefully: zero or more  $a$ 's followed by a single  $b$  followed by the same number of  $c$ 's as  $a$ 's.